

Module 6: Metabolism — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Apply the laws of thermodynamics to biological systems
2. Distinguish between kinetic and potential energy in living organisms
3. Explain how enzymes catalyze reactions by lowering activation energy
4. Describe the structure, function, and recycling of ATP
5. Distinguish anabolic pathways from catabolic pathways with examples
6. Explain oxidation-reduction (redox) reactions and electron carriers
7. Describe feedback inhibition and its role in metabolic regulation

Key Terms to Know

- **Metabolism** — The sum of all chemical reactions in an organism
- **Anabolic Pathway** — Building complex molecules from simpler ones (requires energy input); examples: protein synthesis, photosynthesis
- **Catabolic Pathway** — Breaking complex molecules into simpler ones (releases energy); examples: cellular respiration, digestion
- **Free Energy (ΔG)** — Energy available to do work; negative ΔG = spontaneous (exergonic), positive ΔG = non-spontaneous (endergonic)
- **Exergonic Reaction** — Releases free energy ($\Delta G < 0$); products have less energy than reactants
- **Endergonic Reaction** — Requires free energy input ($\Delta G > 0$); products have more energy than reactants
- **Enzyme** — Biological catalyst (usually a protein) that speeds up reactions without being consumed
- **Substrate** — The specific molecule(s) an enzyme acts upon
- **Active Site** — The region of an enzyme where the substrate binds and the reaction occurs
- **Activation Energy (E_a)** — The minimum energy required to initiate a chemical reaction

- **ATP (Adenosine Triphosphate)** — The primary energy currency of cells
- **ADP (Adenosine Diphosphate)** — The product when ATP releases one phosphate group and energy
- **Cofactor** — A non-protein helper (metal ion or organic molecule) required by some enzymes
- **Coenzyme** — An organic cofactor, often derived from vitamins (e.g., NAD^+ , FAD)
- **NAD^+ /NADH** — Electron carrier; NAD^+ is oxidized form, NADH is reduced form
- **FAD/FADH₂** — Electron carrier; FAD is oxidized form, FADH₂ is reduced form
- **Oxidation** — Loss of electrons (OIL); often involves loss of hydrogen atoms
- **Reduction** — Gain of electrons (RIG); often involves gain of hydrogen atoms
- **Competitive Inhibition** — Inhibitor binds active site, blocking substrate
- **Non-competitive (Allosteric) Inhibition** — Inhibitor binds elsewhere, changing enzyme shape
- **Feedback Inhibition** — End product of a pathway inhibits an earlier enzyme in that pathway
- **Denaturation** — Loss of enzyme 3D shape (and function) due to extreme temperature, pH, etc.

Key Concepts

Laws of Thermodynamics

1. **First Law (Conservation)** — Energy cannot be created or destroyed, only transformed
2. **Second Law (Entropy)** — Every energy transfer increases the overall entropy (disorder) of the universe

Living organisms maintain local order by constantly taking in energy and releasing heat — they are open systems that decrease their own entropy at the cost of increasing entropy in their surroundings.

Energy in Living Systems

- **Kinetic energy** — Energy of motion (e.g., muscle contraction, molecular movement)

- **Potential energy** — Stored energy (e.g., chemical bonds in glucose, concentration gradients)
- Chemical energy in food molecules is a form of **potential energy** that cells convert to **kinetic energy** and **heat**

ATP: The Energy Currency

- **Structure:** Adenine base + ribose sugar + 3 phosphate groups
- **Energy release:** $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ (breaks terminal phosphate bond, releases ~ 7.3 kcal/mol)
- **Energy coupling:** Cells link exergonic reactions (like ATP hydrolysis) to endergonic reactions (like building proteins)
- **ATP cycle:** ATP is continuously recycled — a human uses roughly their body weight in ATP per day

How Enzymes Work

- Lower the activation energy (E_a) without changing the free energy of the reaction
- Are **not consumed** — they are recycled after each reaction
- Are **specific** to their substrate(s) — induced-fit model (enzyme shape adjusts to substrate)
- Affected by: **temperature, pH, substrate concentration, inhibitors**
- Optimal conditions vary by enzyme (e.g., pepsin works best at pH 2, trypsin at pH 8)

Redox Reactions and Electron Carriers

- Redox reactions always occur in pairs: one molecule is **oxidized** (loses electrons), another is **reduced** (gains electrons)
- **Electron carriers** (NAD^+ , FAD) shuttle high-energy electrons between metabolic reactions
- $\text{NAD}^+ + 2\text{H} \rightarrow \text{NADH} + \text{H}^+$ (NAD^+ is reduced to NADH)
- These carriers are essential links between the metabolic pathways you'll study in Module 7

Metabolic Pathway Regulation

- **Feedback inhibition:** The product of a multi-step pathway inhibits an enzyme early in the pathway — this is efficient because it prevents waste
- **Allosteric regulation:** Molecules bind to sites other than the active site, activating or inhibiting the enzyme
- Cells regulate metabolism to match changing needs — they don't run all pathways at full speed simultaneously

The Big Picture: Why Metabolism Matters

Metabolism is the foundation for understanding everything that follows in this course:

- **Module 7** will apply these concepts to the two major energy pathways: photosynthesis (anabolic) and cellular respiration (catabolic) — two sides of the same equation
- **Enzymes, ATP, and electron carriers** ($\text{NAD}^+ / \text{NADH}$, $\text{FAD} / \text{FADH}_2$) are the key molecular players in those pathways
- Every cell in every organism relies on these metabolic principles

Study Tips

1. **Remember OIL RIG** — Oxidation Is Loss (of electrons), Reduction Is Gain
2. **Draw the ATP $\leftrightarrow$ ADP cycle** — Show phosphate release and energy coupling
3. **Distinguish exergonic vs. endergonic** — Think of it as "exit energy" vs. "enter energy"
4. **Know your enzyme factors** — Temperature, pH, concentration, and inhibitor type
5. **Connect to the next module** — Metabolism sets up photosynthesis and respiration; both use ATP, enzymes, and electron carriers
6. **Practice feedback inhibition** — Draw a simple pathway ($A \rightarrow B \rightarrow C \rightarrow D$) and show how D inhibits the $A \rightarrow B$ enzyme